CaveNet: Functional Convergence Under Pressure

Liz Lemma Future Detective

May 24, 2026

Table of Contents

1 I. The Question

In Paper 1, we treated a subjective trajectory as a constructed object: a history-sensitive path through state space shaped by input, retention, valuation, and action. It gave us a vocabulary for describing such paths without reducing them to isolated moments. The present paper begins one step further back. Instead of asking how a given architecture produces a trajectory, we ask why certain transformations of trajectories are repeatedly worth producing.

A useful way to frame the problem is not as a catalog of modules, but as a relation between environmental demand and implementable capacity. Where delayed consequences matter, a state that can carry information across time becomes useful. Where access is limited, mechanisms that preserve relevant information under bottleneck become useful. Where future exposure can be changed, state becomes useful when it can guide sampling or action. Yet none of these pressures by itself guarantees the corresponding function; a system must also have the structural means to instantiate it.

Thus the recurring roles are not treated here as psychological ornaments attached to a preferred model. They are hypotheses about mathematical solutions that become attractive when particular pressures meet particular capacities. This shifts the explanatory burden from Cave as a named architecture to recurrence as a functional pattern.

The guiding question, then, is deliberately comparative. We are not asking why Cave, having been designed with expectation, selection, value retention, regulation, and topology, contains those named roles. That would at most justify an internal taxonomy. We are asking whether those names identify attractors in a larger design space. If a substrate can preserve state, update selectively, couple state to later consequence, and alter its future input stream, then some transformations become more useful than others. Delayed outcomes reward states that carry predictive information. Bottlenecks reward selective preservation. Controllable exposure rewards state-dependent sampling. Structured repetition rewards latent organization that makes later transitions easier to anticipate or control.

This formulation also sets a stricter evidential standard. Resemblance cannot be inferred from vocabulary alone, and it cannot be granted because two systems are drawn with analogous boxes. We should see it in trajectory behavior: convergence under matched pressure, separation under different treatments, collapse when the relevant capacity is removed, and preservation when irrelevant details vary. The claim is therefore modest but general: Cave supplies a reference vocabulary, while pressure and capacity supply the reason to expect comparable functions elsewhere.

The central question can therefore be stated as a bridge between two explanatory levels. Paper 1 supplied a formal vocabulary for describing what a subjective trajectory is and how it may be measured. The present paper asks why certain trajectory transformations should appear in the first place. Our answer is not that a role exists because it has been named, nor that every capable system must reproduce the Cave architecture. Rather, we treat each role as a candidate solution to a recurrent problem: when a pressure makes some transformation useful, and a substrate has the capacity to implement it, that transformation becomes a plausible target of adaptation, design, or learning.

This gives the paper its basic test. If the explanation is correct, then matched pressures should organize populations of trajectories in recognizable ways, while removing the relevant capacity should disrupt that organization. Conversely, if a system faces the pressure but lacks recurrence, selective update, state–value coupling, or exposure control, the corresponding function should not appear merely by stipulation. Negative cases are therefore not exceptions to the thesis; they are part of its content. The point is to move from a vocabulary of roles to an account of why those roles recur, when they fail, and what kind of resemblance can be claimed across different substrates.

2 II. The Thesis

We can state the central hypothesis compactly: capacity + pressure $\rightarrow$ useful mathematical function. By this we do not mean that pressure alone creates structure, or that capacity alone determines its use. Pressure supplies the reason a transformation matters. Capacity supplies the machinery by which such a transformation can be realized. A world may reward delayed use of prior information, but a system without any way to retain state cannot exploit that delay. Conversely, a system may possess rich internal state, but without a recurring problem that makes some transformation useful, that capacity need not organize into the relevant function.

The thesis is therefore comparative rather than merely taxonomic. We are not treating the Cave roles as arbitrary named modules and then asking whether other systems contain modules with the same labels. We are asking whether, under matched pressures, systems with the required capacities tend to develop transformations that play similar mathematical roles. The question is functional: what problem is being solved, what capacity makes the solution possible, and what measurable change in trajectory geometry results?

The reference vocabulary gives us a disciplined way to pose that question. Each role names a pressure-capacity pairing and the kind of trajectory effect that would count as resemblance.

We can read the five reference roles in this way. First, temporal recurrence together with state retention supports expectation: prior state becomes usable as a predictor of later input, and successful organization is visible as reduced prediction error or improved future-state recovery. Second, limited access together with selective update supports attention-like selection: when not all inputs can be preserved or acted upon, relevant input is preferentially admitted into the trajectory while irrelevant variation is allowed to fall away. Third, delayed value together with state/value coupling supports value-shaped retention: states that matter for later consequence are held more reliably than neutral states. Fourth, future coupling together with action or exposure output supports regulation: the present state changes later sampling, exposure, or action, and thereby alters the future stream to which the system is subject. Fifth, structured repetition together with latent geometry supports topology-like organization: repeated experience induces stable regions, characteristic transitions, and preserved relations among histories rather than merely a list of isolated events.

This qualification is important because it makes the claim falsifiable. A pressure can be present in the environment while the relevant function is absent from the system. Delayed consequence, for example, does not by itself produce memory; it only makes memory valuable to an architecture able to carry information across the delay. Likewise, a bottleneck does not by itself produce selection; it creates a condition under which selective updating would be useful if the system can differentially preserve some inputs over others. The converse failure is equally important. A system may have recurrence, hidden state, or high-dimensional latent variables, yet if no recurring demand favors prediction, retention, allocation, or regulation, those degrees of freedom may remain unused, drift into irrelevant encodings, or support some other task.

Thus the empirical test must include both positive and negative cases. When pressure and capacity are jointly present, we ask whether population trajectories acquire the predicted deformation. When one side is removed, weakened, or mismatched, we should expect the corresponding structure to collapse, degrade, or fail to appear. The theory therefore treats absence as informative: a missing function under a missing capacity is not a counterexample, but part of the map of when role emergence should and should not occur.

3 III. Reference Functions

We now fix what is meant by a reference function. The term is deliberately narrower than a psychological faculty and broader than a particular circuit. A reference function is an operational pattern in the evolution of a state trajectory: given a class of inputs, starts, and constraints, it specifies what information is preserved, transformed, suppressed, or made predictive over time. Thus the object of comparison is not whether a system contains a component bearing the same name, but whether its trajectories exhibit the same measurable deformation under matched pressure.

This distinction matters because the same role may be implemented by very different substrates, and because a substrate may fail to implement it even when the world would make it useful. We therefore treat the Cave vocabulary as a coordinate system for measurement rather than as an inventory of parts. Expectation, selection, value retention, regulation, and topology name candidate transformations of trajectories. They are tested by convergence, separability, decodability, ablation, and control failure. If these signatures appear, we have evidence for role resemblance; if they disappear when the enabling capacity is removed, we have evidence about the mechanism that carried the role.

The reference list below should therefore be read as a set of operational targets.

Pressure	**Required capacity**	**Reference function**	**Trajectory effect**
Temporal delay	recurrence and state retention	expectation	prior state predicts later input
Limited access	selective update	selection	relevant input is preserved under bottleneck
Delayed consequence	state–value coupling	value retention	future-relevant state persists
Future coupling	action or exposure output	regulation	state alters later sampling or input
Structured repetition	latent geometry	topology	trajectories form stable regions and transitions

The logic of the table is conjunctive. A pressure explains why a transformation would be useful, but it does not by itself create that transformation. A capacity explains how such a transformation could be implemented, but it does not by itself determine that the transformation will be used. We therefore read each row as a claim of the form

$$\text{capacity} + \text{pressure} \rightarrow \text{useful mathematical function.}$$

Temporal recurrence under delayed input makes expectation useful because retained state can reduce later uncertainty. Selective update under limited access makes attention-like selection useful because some inputs must be preserved at the expense of others. State–value coupling under delayed consequence makes value-shaped retention useful because neutral and future-relevant states should not decay equivalently. Output control under future coupling makes regulation useful because present state can change later exposure. Latent geometry under structured repetition makes topology-like organization useful because recurrent trajectories can settle into regions and transitions.

The table is therefore a reference basis, not a catalog of faculties that we assume to be present. It fixes the meaning of resemblance before we ask whether resemblance appears. To say that a new substrate exhibits expectation-like behavior is not to say that it contains a named expectation module; it is to say that, under delay and with recurrence, earlier state becomes measurably informative about later input and reduces uncertainty. Likewise, to say that a system exhibits value retention is to say that future-relevant state is preferentially preserved under delayed consequence, not that the system has inherited the Cave value coordinate.

This restriction is important for both positive and negative evidence. Positive evidence consists in trajectory effects: convergence, separability, preserved decodability, altered path length, and stable regions of state–space organization. Negative evidence consists in their disappearance when the enabling capacity is removed or when the pressure is neutralized. A substrate with no recurrence may face delayed consequence, but it cannot carry the relevant state. A substrate with recurrence but no appropriate pressure may have capacity without adopting the function.

Thus the reference functions are neither arbitrary labels nor guaranteed discoveries. They are operational targets. They tell us what would count as the same mathematical role appearing elsewhere, while leaving open whether any particular system can in fact realize it.

4 IV. Capacity As Precondition

The thesis of this section is deliberately double. We are not saying that a pressure, merely by existing in an environment, summons the corresponding role into being. A delayed consequence may make memory useful, but it does not create memory in a mechanism that has no way to preserve state across time. If the current observation is the only variable available to the system, then events separated by a delay remain mathematically disconnected for that system. Likewise, a bottleneck may make selection valuable, but it cannot yield attention-like allocation unless the substrate has some means of updating one part of its state while protecting another, or of weighting some inputs over others.

Thus the formula must be read in both directions: pressure supplies the reason a function would matter, while capacity supplies the degrees of freedom through which that function could be implemented. The absence of either side changes the expected result. With capacity but no pressure, a mechanism may remain unused, arbitrary, or weakly organized. With pressure but no capacity, the system may face the right problem yet be unable to form the relevant transformation. We therefore treat capacity not as an embellishment, but as a boundary condition on emergence.

For this reason, we count certain failures as positive information rather than as mere absence of effect. A control that faces the same delayed outcome but has its hidden state reset should not retain delayed information. A non-recurrent controller in the same world should not regulate future exposure when the present observation is locally ambiguous. A bottlenecked system with no selective update should not preserve task-relevant input better than chance. In each case, the pressure is still present in the environment, but the relevant degree of freedom has been removed from the mechanism.

The evidential pattern is therefore comparative. If the capable condition develops a trajectory transformation and the matched incapacity control does not, the difference does not simply show that the first system performed better. It localizes the source of the function. It tells us that the observed role depends on a carrier: recurrence for delayed state, selective update for bottleneck allocation, state–value coupling for future-relevant retention, action or exposure output for regulation. These negative cases keep the argument from becoming circular. We do not infer a role merely because a task would benefit from it; we ask whether the role disappears when the capacity required to implement it is removed.

This point also fixes the interpretation of null results. A failed role recovery is not automatically a failed experiment; it may be the predicted consequence of removing the relevant carrier. When the carrier is absent, the appropriate conclusion is not that the pressure was weak or that the role was ill-defined. It is that the mathematical route from problem to function has been closed. The environment may still reward delayed use of information, differential preservation, or future regulation, but the mechanism has no coordinate along which that reward can be converted into a stable trajectory transform.

In this framing, capacities are not guaranteed modules; they are enabling structures. Recurrence, selective update, coupling to value, and controllable output specify families of admissible transformations. Pressure then selects among those admissible transformations by making some of them useful. This is why our tests must include matched incapacity conditions. They are the guardrail against a teleological reading of the argument. We do not claim that systems discover expectation, selection, value retention, regulation, or topology because these names are available in our vocabulary. We claim that, when the relevant degrees of freedom exist and the world imposes the corresponding pressure, population trajectories should reorganize toward the matching mathematical role. Where those degrees of freedom are absent, the reorganization should fail. Pressure is not magic; it is selection over implementable possibilities.

5 V. The Evaluation Contract

We evaluate every candidate system through the same source-neutral contract. An experience protocol is first presented to a subject or substrate, whether that substrate is the authored Cave model, a weakened CaveNet variant, a learned controller, an evolved recurrent subject, or a negative control. The substrate then produces an Episode. The Episode is not treated as an interpretation by itself; it is the shared measurement surface from which trajectory readouts are computed. Those readouts are then compared across populations of runs, aligned by treatment, start condition, and matched set, and finally tested against controls that remove, weaken, or scramble the relevant capacity.

In schematic form, the contract is

$$\begin{aligned}&\text{experience} \to \text{subject or substrate} \to \text{Episode}\\ &\qquad \to \text{trajectory readouts} \to \text{population comparison} \to \text{controls.}\end{aligned}$$

This order matters. We do not begin by asking whether an internal coordinate has the same meaning across architectures. We ask whether the same pressure, when passed through a capable system and measured through a common interface, produces comparable trajectory transformations. The Episode therefore functions as an evidential adapter: it lets heterogeneous systems enter the same evaluation pipeline without pretending that their internal variables are identical. Only after this translation do we assign functional claims.

The common surface is a typed record of what the protocol made available, what the system anticipated, what it retained, and how it acted. At each relevant time step it stores the actual input x_t and, when available or reconstructable, an expected input $\hat{x}_t$. It also stores the carried state: this may be an authored memory trace, a CaveNet state variable, or a learned hidden vector. We do not require these states to share coordinates; we require that they be available for the same downstream tests.

The Episode also records quantities that connect state to function. Surprise or prediction error marks the gap between expected and actual input. Learning-rate or update-gain fields indicate how strongly new evidence changes the carried state. Attention, selection, or exposure variables record which inputs are privileged, suppressed, approached, or avoided. Active-input fields identify what the subject actually sampled or made consequential, rather than merely what the world offered. Finally, metadata records treatment identity, start condition, matched-set membership, condition, seed, and control role. These labels are not ancillary bookkeeping. They are what allow trajectories to be aligned, separated, permuted, ablated, and compared as populations rather than as isolated runs.

Internally, these systems are deliberately non-isomorphic. In the reference Cave model, expectation, selection, value retention, regulation, and topology are named components of the update rule. In CaveNet, related operations are mediated by parameterized gains and network state, so the relevant question is whether changing or learning those gains moves the Episode readouts in the predicted direction. In minimal or weakened subjects, some machinery is absent by design; their failures are informative because they show which capacities the pressure cannot replace. In the evolved recurrent subjects, the contrast is sharper still: the system has only observations, a recurrent hidden state, and an exposure output. It has no native expectation variable, no authored value trace, and no declared Cave topology.

For that reason, our evaluation contract does not license coordinate-by-coordinate identification. A Cave feature axis and a learned hidden dimension may both support future-outcome decodability, but they are not thereby the same representational object. The claim is weaker and, for our purposes, more testable: under matched pressure, systems with the required capacity should exhibit similar transformations in trajectory geometry and similar collapses when that capacity is removed or scrambled. The contract therefore treats architectural diversity as the test bed for functional resemblance, not as an obstacle to evaluation.

6 VI. Population Trajectory Geometry

We therefore shift the evidential unit from role labels to geometry. A single run can always be narrated after the fact: a variable can be called expectation, a hidden coordinate can be called value, and a visual cluster can be called a state. Those names become informative only when they predict how the run will move under intervention. If a pressure is shared, initially different starts should bend toward a common region, while still retaining some history of where they began. If a start is held fixed and the pressure changes, the resulting paths should separate in a treatment-specific direction. If the relevant capacity is removed, the same separation should collapse; if the capacity is restored or adapted, it should reappear. Likewise, if temporal order or value structure is permuted, the corresponding deformation should weaken in a way specified before inspection.

This is why the population view matters. We are not asking whether a single trace resembles a familiar diagram. We are asking whether families of traces exhibit reproducible contraction, divergence, preservation, and failure modes under matched comparisons. The claim is geometric before it is semantic: a proposed role earns its name by producing the trajectory transformations that the role mathematically requires.

For this reason, each element in the population is not merely a trajectory, but a trajectory indexed by a small design grammar. The treatment specifies the shared input sequence or pressure protocol. The start condition records the initial state, seed, subject, or prior from which the run begins. The mechanism condition identifies the substrate or variant that produced the path: reference system, ablated system, adaptive controller, recurrent subject, shuffled-order control, and so on. The comparison role states how the run is to be used inferentially—as baseline, treatment, negative control, or intervention control. Finally, the matched set identifies runs that should be aligned before comparison, for example the same initial seed across treatments, or the same pressure protocol across mechanisms.

This indexing is deliberately modest, but it prevents a common ambiguity. Without it, separation may be confused with different starts, convergence may be confused with averaging, and a control failure may be mistaken for noise. With it, the geometry becomes conditional: we can ask what changes when treatment varies while start is held fixed, what persists when start varies while treatment is shared, and what disappears when the relevant mechanism is removed.

The resulting measurements are correspondingly relational. We first ask whether starts exposed to the same treatment contract over time: pairwise distances among matched starts should decrease more than expected from the initial spread, without necessarily reaching zero. We then ask the complementary question: when the same start is carried through different treatments, do the paths separate in a treatment-specific direction? This same-start treatment separation is the basic test that experience has deformed the trajectory rather than merely revealed pre-existing differences.

Path length gives a related but distinct quantity. A pressure may bring runs to a similar endpoint by a short correction or by a long excursion through latent space; these are not the same transformation. Final spread records how much heterogeneity remains after the protocol. Future-outcome decodability asks whether the state at time t contains information about later consequences, not merely about the current observation. Control-versus-treatment separation asks whether the geometry produced by the intact mechanism survives comparison to ablated or weakened variants. Finally, permutation-null separability tests whether the apparent treatment structure exceeds what would be obtained after shuffling the relevant temporal, value, or alignment labels.

Together, these metrics turn visual impressions into falsifiable population claims. A role-like function should not only make a plot look organized; it should produce the predicted pattern of contraction, separation, retained information, and collapse under the appropriate controls.

7 VII. Population Designs

We begin with two complementary population designs because a single trajectory cannot distinguish imprinting, convergence, and collapse. In the first design we hold treatment and mechanism fixed while varying the start: one treatment $\times$ many starts $\times$ one condition. The question is not simply whether the treatment has an effect. It is whether a shared pressure bends initially distinct subjective states toward a common region of trajectory space while still preserving enough history to identify where each run came from.

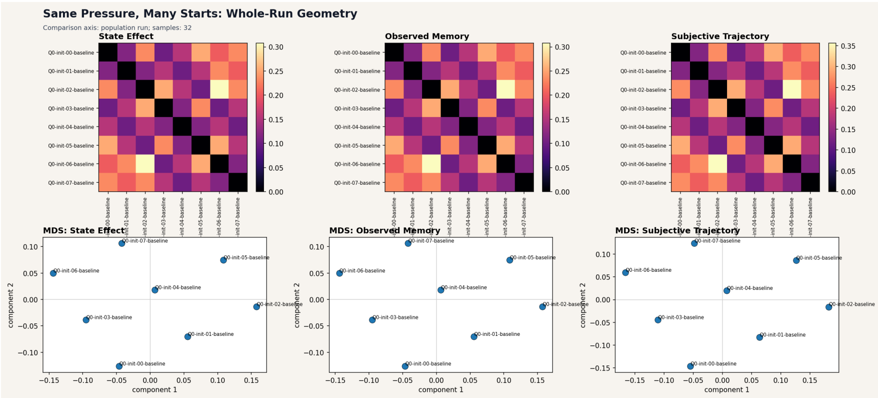

Figure 1: Same-pressure, many-starts design. Shared pressure should deform initially different trajectories toward common structure without reducing them to a single copied endpoint.

This design therefore looks for contraction without erasure. If every start remains scattered, then the treatment has not organized the population. If every start becomes identical, then the medium is acting like a reset or copy operation rather than a trajectory-transforming subject. The evidential pattern lies between these extremes: distances among starts should decrease, curvature should become partially shared, and final states should occupy a structured basin rather than a single point. We can measure this as within-treatment start convergence over time, final spread, path length, and persistence of start-specific residual structure. In this sense the first design tests whether pressure can induce common organization without abolishing initial condition.

Second, we reverse the comparison: many treatments $\times$ matched starts $\times$ one condition. Here the initial state, seed, subject, or prior is aligned across treatments, and the mechanism is held fixed. What varies is the experienced sequence or pressure protocol. The question is whether treatment identity leaves a recoverable deformation in trajectory space, rather than merely producing a different final report or a different plotted image.

The expected signature has two parts. For a matched start, different treatments should push the run through distinguishable regions, with separable curvature, path length, endpoint distribution, or future-outcome decodability. For a fixed treatment, different matched starts should show related bending, so that the treatment contributes a common component to the trajectory even when individual history remains visible. Thus the evidence is not simple divergence and not simple convergence. It is an interaction: starts explain some residual variation, treatments explain a reproducible deformation, and the condition determines whether the substrate can carry that deformation.

Operationally, we test this with treatment separability against permutation nulls, same-start treatment separation over time, and preservation or collapse under mechanism controls. If labels remain recoverable only when the relevant capacity is present, then the population geometry is carrying treatment-shaped structure.

This is why the second design has a special status in the argument. Paper 1 followed the formation of a single subjective trajectory: how prior state, incoming episode, update rule, and role readout produce a path through the subject's space. The present paper asks a higher-order question. Once such paths exist, do families of them reorganize in lawful ways when pressure is varied?

The matched-start design is the hinge between these questions. It lets us lift the analysis from within-run construction to between-run geometry. If the same start, placed under different treatments, moves through separable paths, then treatment is acting as a deformation of trajectory space. If different starts under the same treatment acquire related curvature, then that deformation is not an idiosyncratic accident of one run. It is a population-level signature.

This transition matters because role emergence is not established by naming a hidden state after a reference role. It is established when a capable system, under matched pressure, comes to preserve, separate, contract, or regulate trajectories in the way that role mathematically requires. Thus design two preserves the central lesson of Paper 1, that subjectivity is dynamically produced, while adding the central demand of Paper 2: produced trajectories must become comparable as a population under pressure and control.

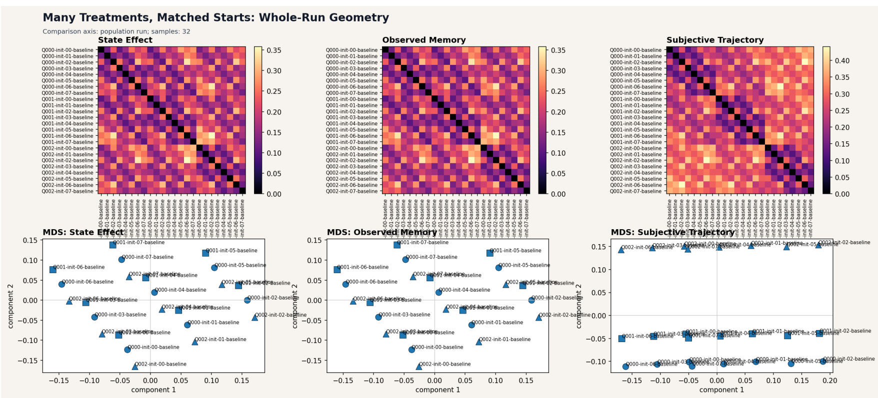

Figure 2: Many-treatment, matched-starts design. Holding starts fixed lets treatment-specific deformation be read as trajectory effect rather than preexisting difference.

8 VIII. Conditions And Control Failures

We therefore treat mechanism variants as geometric probes rather than as mere implementation details. A variant asks what happens to a population of trajectories when one candidate capacity is weakened, removed, or redirected while the pressure protocol is held fixed. If the corresponding role was carrying treatment-relevant structure, the matched population should not merely become noisier; it should lose a specific form of separability, convergence, decodability, or future coupling. Conversely, if the geometry is preserved, then the claimed role was either not necessary for that effect or was redundantly supported elsewhere in the system.

This is why failures under control conditions are evidential rather than incidental. They mark the boundary between pressure and capacity. The same external sequence can continue to impose delayed consequence, bottleneck, or structured repetition, but a substrate whose relevant channel has been removed may no longer be able to express the trajectory transform. In that sense, the control does not deny the pressure; it tests whether the proposed mechanism is the route by which the pressure becomes mathematically useful. The following variants should be read in that spirit: each removes or isolates one access path, and the resulting population geometry tells us which capacities actually carry the recovered role.

The zero-attention variant is the bluntest probe. It removes the deployed selective channel, so that treatment access is reduced to chance or uniform sampling. If the treatment protocol still differs externally but the population trajectories no longer separate by treatment, the loss is informative: the pressure remains present, but the system can no longer route the relevant differences into its state update.

The external-only variant asks a narrower question. Here the current actual input still contains the treatment-specific information, but the expectation readout is removed. We should therefore not expect a collapse of all treatment effects. Effects carried directly by present input may remain. What should disappear is the component of the geometry that depends on comparing the present against retained prior structure. Conversely, the internal-only variant preserves access to prior state while collapsing treatment access through current actual input. This isolates whether the apparent role was being carried by memory-like continuation rather than by immediate sensory access.

Finally, the no-memory condition preserves treatment variation in the current stream while preventing treatment recovery from observed memory. It is especially useful for distinguishing transient input sensitivity from retained trajectory structure. A system may still react locally, but it should no longer support the same delayed recovery of treatment identity from its state history.

The zero-prediction condition completes this diagnostic pattern. It removes the explicit expectation objective without removing actual input variation or memory variation. If treatment separability survives there, the result is not a contradiction. It tells us that separability was carried by input and retained state rather than by the prediction channel itself. In that case the control is an expectation-role control, not an access-control condition.

Taken together, these variants make the role vocabulary operational. A role is not confirmed because we can name it after inspecting a run. It is confirmed, weakened, or rejected by the way population geometry changes when the corresponding capacity is intervened on. We ask whether matched starts still diverge by treatment, whether within-treatment trajectories still converge, whether future outcomes remain decodable, and whether the relevant collapse is selective rather than global. The important signature is not mere performance loss, but patterned loss: the disappearance of precisely the structure that the proposed role was supposed to carry.

This also gives the vocabulary its executable character. Functions can be removed, partially preserved, restored, or bypassed, and each case leaves a measurable trace in trajectory geometry. Control failure is therefore not an embarrassment to the framework. It is one of the main ways the framework identifies which capacities make pressure mathematically consequential.

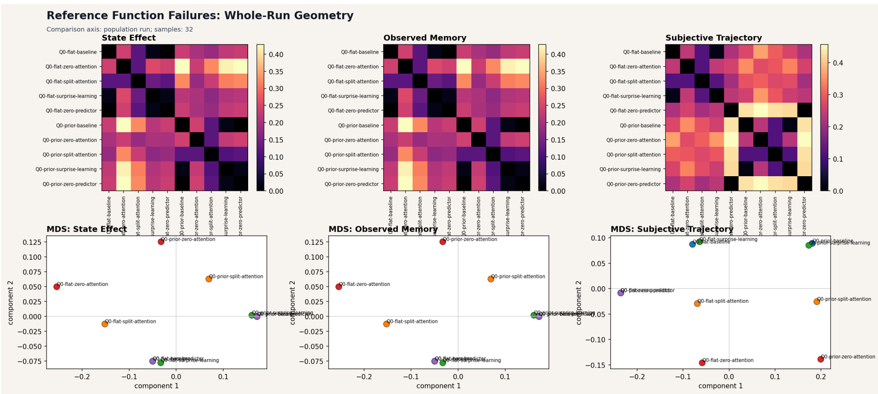

Figure 3: Reference-function control failures. The evidential target is patterned loss: treatment geometry weakens or collapses when the proposed carrier is removed or isolated.

9 IX. Pressure-Shaped CaveNet

CaveNet occupies a deliberately intermediate position in the argument. It is not introduced as a blank organism in which functions appear from nowhere. Rather, it is a network-form implementation of the Cave update: a system whose variables, gates, and readouts are already organized in the vocabulary of expectation, selection, value retention, regulation, and topology. For that reason, success in CaveNet cannot by itself establish open-ended role emergence. The architecture has been given much of the right language in advance.

That limitation is also why CaveNet is useful. Before asking whether distant substrates discover analogous functions, we first need to know whether pressure signals can move a weakened, role-compatible system in the predicted direction. CaveNet therefore serves as a calibration instrument. It asks a modest but important question: when the required capacities are present but degraded, does pressure reshape the update toward the reference behavior, or does the role vocabulary remain merely descriptive?

This section should be read in that spirit. A positive result here is a compatibility result, not an origin story. A negative result would be equally informative, because it would show that even an architecture designed to express the roles does not reliably improve under the proposed pressures. CaveNet thus anchors the bridge between operational reference functions and stronger emergence tests.

The pressure-shaped population report therefore asks for relative movement, not for spontaneous invention. For each generated world we hold the treatment structure fixed and compare three anchors: a weakened CaveNet with fixed gains, a weakened CaveNet whose gains are allowed to adapt to pressure signals, and the fixed reference CaveNet. The relevant question is whether the weak system moves closer to the reference trajectory geometry when pressure is supplied, and whether that movement is stable across worlds rather than confined to a favorable example.

We also train a controller that maps pressure summaries into gain settings. This controller is not a new substrate in the strong sense; it still acts through the CaveNet interface. Its value is diagnostic. If it generalizes to held-out generated worlds, then the pressure variables contain usable information about how the update should be reshaped. If ablations remove particular pressure inputs and selectively reduce the gains, then the mapping is not merely a generic optimization trick. It identifies which pressure channels support which improvements.

Thus the comparison has two parts: a fixed pressure rule tests whether a hand-specified deformation is directionally adequate, while the learned controller tests whether pressure-to-gain structure can be recovered from data.

The conclusion is therefore deliberately modest. The fixed pressure rule shows that the proposed pressures are not inert: in many generated worlds they nudge the weakened update in the reference direction. But the effect is small and heterogeneous. Some worlds improve, some barely move, and the aggregate displacement is not large enough to treat the rule as a general recovery procedure. At most, it says that a hand-written pressure deformation is partially aligned with the reference geometry.

The learned controller gives a stronger calibration result. When pres-

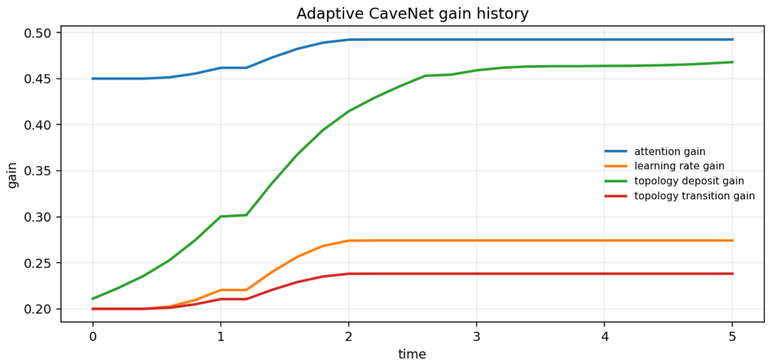

Figure 4: CaveNet gain history under pressure. The relevant claim is calibrated adaptation within a role-compatible architecture, not spontaneous role invention.

sure summaries are allowed to determine gain settings, movement toward the reference becomes larger and more consistent across held-out worlds. The ablation pattern also matters: removing particular pressure channels selectively weakens the corresponding improvement, which indicates that the controller is using structured pressure information rather than merely fitting an average correction.

Even so, this remains a pressure-shaped compatibility result. CaveNet already contains the representational slots through which expectation, selection, retention, regulation, and topology can be expressed. The experiments therefore show that pressure can tune a weakened role-compatible mechanism toward those functions; they do not show that an unstructured substrate invents the functions. CaveNet supports the bridge, but it is not the far side of the bridge.

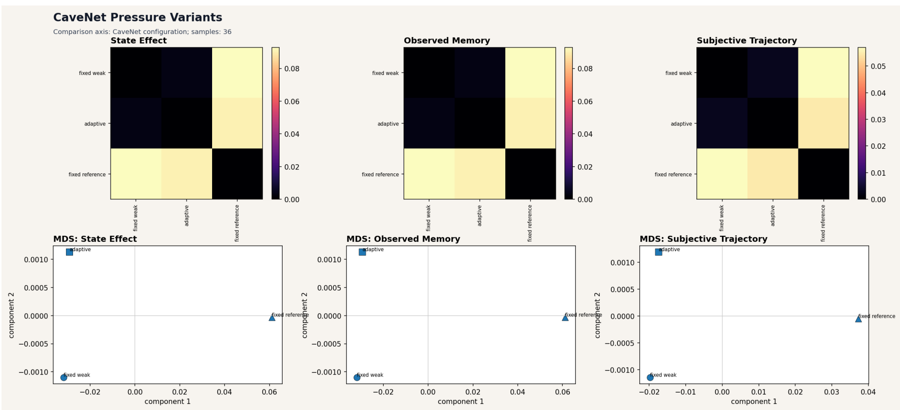

Figure 5: Pressure-shaped CaveNet dashboard. Weak, adaptive, and reference configurations are compared through the shared episode surface.

10 X. The Evolved Recurrent Subject

We now leave the reference architecture behind. The relevant subject is deliberately small: no hand-designed role inventory, no labeled surfaces through which we can read Cave variables. Its state update is

$$h_t = \tanh(W_x \, \mathrm{obs}_t + W_h h_{t-1} + b_h), \qquad \mathrm{exposure}_t = \sigma(W_a h_t + b_a).$$

This is enough recurrence to let past observations influence later action, and enough output coupling for hidden state to alter future sampling. It is not enough, by construction, to smuggle in the reference explanation. We therefore treat it as the strongest current non-reference test in the paper: if role-like structure appears here, it cannot be attributed to authored Cave coordinates or to an explicit module bearing the relevant name.

The interpretive question is consequently narrow and empirical. We ask whether optimization can make the hidden trajectory carry information that is useful across a delay, and whether that information is deployed in exposure. The unit of comparison is not an axis-by-axis match to the reference model, but a causal pattern: retained state, recoverable future-outcome information, and action that changes subsequent input exposure.

The only thing this subject is evolved to do is manage exposure in a delayed cue–outcome world. Its fitness does not ask it to predict the next observation, reconstruct the past, label a cue as valuable, or organize experience into a Cave-like map. The environment supplies pressure, but the architecture supplies only recurrence and an exposure output. Thus any useful internal structure must be purchased indirectly through performance: if retaining a cue helps later exposure, selection can favor retention; if it does not, there is no auxiliary objective that preserves it for our convenience.

This matters because the negative specification is part of the test. There is no native expectation variable whose accuracy we train. There is no prediction loss that would force hidden state to encode future input. There is no explicit memory trace that can be inspected as the subject's remembered cue. There is no authored topology layer and no module named value retention. At most, the controller can learn a latent trajectory in which earlier observations continue to shape later exposure decisions. By design, then, the experiment begins without the reference roles installed; it asks whether the pressure/capacity pairing is sufficient to make some of their mathematical functions appear.

What we observe is therefore the predicted causal shape. Under delayed value and exposure-control pressure, evolution favors hidden trajectories from which the later outcome can be decoded before it is observed, and those same trajectories drive differentiated exposure. The hidden state is not merely a by-product of sensory history; it becomes an action-relevant carrier of information across the cue–outcome gap. This is the sense in which the subject acquires a value-retention and regulation function: not because a labeled variable appears, but because past cue information remains causally available when exposure must be set.

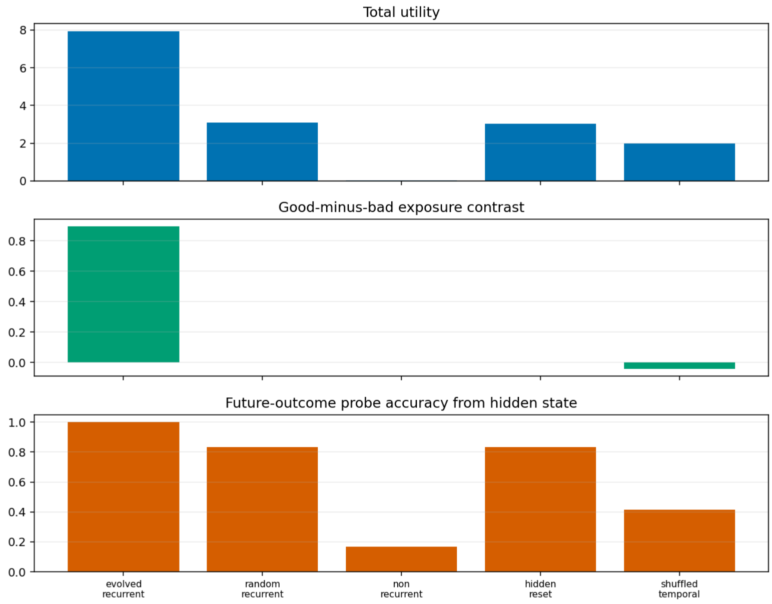

Figure 6: Evolved recurrent subject metrics. Future-outcome information in hidden state and exposure contrast depend on recurrence and temporal order.

The controls identify the dependence on capacity and temporal structure. If we reset hidden state across the delay, the exposure contrast collapses, showing that current observation alone is insufficient. If we remove recurrence altogether, the controller cannot solve the task, even though it faces the same environmental pressure. If we shuffle the temporal order, we weaken both exposure contrast and future-outcome decodability, indicating that the learned state depends on ordered trajectories rather than static cue frequencies. The result is not a coordinate match to the Cave system. It is a functional resemblance: delayed value pressure, given recurrence and action coupling, selects latent state that carries future-relevant information and uses it to regulate future exposure.

11 XI. The Harder World

The first evolved-subject world gives us an important positive result, but it is not yet a decisive one. In that environment, the cue that matters for later outcome is also easy to track by simpler means. A controller can therefore look competent for the wrong reason. It may not be retaining value-bearing state over a delay; it may simply be counting which cue appears most often, or reacting to the most recent salient observation. Such policies could generate exposure contrast, and might even leave decodable structure in hidden state, without solving the intended value-retention problem.

This matters because our claim is functional, not verbal. We cannot infer value-shaped retention merely because behavior resembles value-shaped retention in a world where value is confounded with surface statistics. A stronger test must make the cheap summaries point away from the correct solution. Frequency should no longer identify what matters. Recency should be unreliable because intervening observations occupy the delay. Mere signedness should not be enough to distinguish real consequence from irrelevant signal. Only under that stricter arrangement can we ask whether the recurrent subject is carrying information because it is useful for delayed regulation, rather than because the environment has made a shallow proxy accidentally sufficient.

The dissociation world implements exactly this separation. We assign true delayed consequence to a rare cue, while allowing a common cue to carry the same superficial sign without carrying the consequential outcome. Between cue and outcome we insert distractors, so that the observation closest in time to the regulatory decision is usually not the observation that matters. The environment is therefore adversarial to the simple summaries that the easier world could not exclude.

Under this design, a frequency policy is pulled toward the common cue, precisely because the common cue dominates experience; but that cue is not the one whose delayed consequence should govern exposure. A recency policy is pulled toward intervening distractors, because the relevant cue is no longer the last salient item available before action. A merely signed policy also becomes ambiguous, because signed surface evidence is present both where it matters and where it does not. The correct solution must instead preserve information about the rare, outcome-bearing cue across the delay and use that retained state to shape later exposure.

This is why the dissociation world is a sharper test. It does not merely ask whether the controller can produce contrast. It asks whether contrast survives when the cheap routes to contrast have been made systematically misleading.

The answer is yes, but with an important qualification. The recurrent subject still develops a delayed regulatory state under the dissociation, and it still outperforms the frequency-based alternative. However, the function that emerges is not best described as full magnitude-sensitive value retention. It is more accurately described as sign-gated retention. The controller preserves enough information to distinguish whether a cue predicts a good or bad direction for later exposure, and it uses that distinction to regulate action across the delay. It does not, with equal clarity, separate high-value from low-value consequences.

This precision matters. The learned policy acts on the common signed cue almost as strongly as it acts on the rare consequential cue. In behavioral terms, that can still be useful: when negative outcomes are possible, broad contrast based on sign can reduce losses even if the magnitude estimate is coarse. But as an interpretation of the internal function, we should not call this magnitude selectivity. The harder world therefore strengthens the emergence claim while narrowing it. It rules out frequency and recency explanations, shows that delayed sign information is retained for regulation, and names the recovered role more exactly than the aligned world could: predicted-sign retention with sign-gated exposure, not complete value-magnitude representation.

12 XII. Function Resemblance, Not Coordinate Identity

Before we can ask whether a learned system has acquired a Cave-like role, we must specify what is allowed to count as evidence. The danger is not merely terminological. If we compare internal variables by name, index, or visual similarity, we silently assume the very equivalence we are trying to test. A recurrent controller may distribute a predictive or valuational quantity across many coordinates; another may rotate the same information into a different basis; a third may use transient curvature in its state trajectory rather than a stable scalar channel. These differences matter for interpretation, but they need not matter for the mathematical role performed.

Our boundary is therefore functional rather than anatomical. We ask whether a pressure produces comparable transformations of trajectories: whether uncertainty is reduced when prior state is available, whether delayed outcomes are decodable from the evolving state, whether future exposure is changed by that state, and whether removing the relevant capacity collapses the effect. Such tests preserve the comparison while avoiding a false identification of internal coordinates. They let us say that two systems solve a similar problem in a similar dynamical sense, without saying that they contain the same variables or decompose the solution in the same way.

In the reference Cave model, a feature coordinate is part of the authored description of the subject. When we name an axis as expectation, value, or salience, that name belongs to the construction: it tells us how the update rule is written and what intervention on that coordinate is supposed to do. The coordinate is therefore interpretable by design. Its semantics are not discovered after the fact; they are specified with the model.

An evolved recurrent subject is different. Its hidden vector is only a state space made available to learning or selection. A coordinate h_i may participate in many computations at once, and a computation that looks unitary at the behavioral level may be spread across several coordinates, timing relations, or nonlinear combinations. Conversely, a single unit may contribute to prediction in one region of state space and to action selection in another. For that reason, we do not read a learned hidden dimension as if it were an authored Cave axis. We may decode quantities from it, perturb it, or ask how it affects the trajectory, but those operations do not license the stronger claim that the coordinate itself has the same meaning as the named reference variable.

Accordingly, the evidential unit is a pattern of counterfactual performance across measurements, not a one-to-one map between axes. If access to prior state lowers prediction error, or surprise, we may speak of expectation-like function. If later states or outcomes can be decoded from hidden state during the delay, and if that information is preferentially preserved when neutral information is not, we may speak of value-shaped retention. If changing or resetting that state changes later exposure, we may speak of regulation. If matched starts subjected to different treatments separate, while different starts under the same treatment acquire shared curvature, we may speak of treatment-dependent trajectory geometry. These claims become stronger when they are accompanied by path-length changes in the predicted direction, reduced final spread where convergence is expected, and permutation-null separability where treatment identity should be recoverable.

The crucial tests are destructive as well as constructive. A proposed role should weaken under ablation, temporal shuffle, hidden-state reset, or another matched control that removes the relevant capacity while preserving irrelevant surface statistics. Collapse under the right control is not a nuisance result; it is part of the identification. What survives such tests is a functional resemblance claim: the learned system implements a transformation with the same mathematical use. What does not follow is coordinate identity.

13 XIII. Sweeps And Stability

A single successful evolution run is useful, but it is not yet a stability result. Evolutionary search is stochastic: an apparently clean subject may be a favorable seed, an unusually convenient initialization, or a lucky path through mutation and selection. For that reason, we treat sweeps as the test that separates an illustrative case from a pressure-shaped tendency.

The sweep question is therefore not whether every evolved subject uses the same hidden coordinate system. We should not expect that. Compact recurrent controllers can rotate, distribute, or entangle information across hidden dimensions while preserving the same input-output function. The more appropriate question is whether the same functional signatures recur across independently evolved subjects: future-outcome information in state, exposure contrast under delayed consequence, collapse under memory-disrupting controls, and separable latent geometry under matched protocols.

This is also where negative and partial results become informative. If only one seed exhibits the effect, then the emergence claim is weak. If many seeds exhibit the effect but with different magnitudes, then the claim becomes graded. If a role appears reliably for value retention and regulation but only weakly for selection, the sweep should preserve that distinction rather than average it away. With that standard in place, the sweep results tell us which trajectory transformations are stable products of the pressure and which remain incomplete.

Across the evolved exposure sweep and the evolved role sweep, the recurring pattern is not uniform emergence but graded emergence. The strongest signatures appear where the pressure directly rewards carrying delayed consequence forward in state. Independently evolved controllers repeatedly place future-outcome information in their recurrent dynamics, use that information to modulate exposure, and lose the relevant contrast when recurrence or temporal structure is disrupted. Their latent trajectories also tend to organize by value-relevant structure, especially by outcome sign, rather than remaining an undifferentiated cloud of hidden activations.

The weaker result is selection. The subjects do show non-random sensitivity to cue-bearing input, and in that limited sense the sweep supports cue-weight concentration: some observations matter more than others for the later hidden state and exposure policy. But this is not yet full dynamic attention. We do not see a clean recovered mechanism that allocates access under a bottleneck, updates selectively as future relevance changes, or reconstructs an internal expectation channel comparable to the reference role. Thus the selection result should be named conservatively. The pressure reliably shapes value retention, regulation, and latent value geometry; it produces only a partial selection-like signature. This is exactly the distinction the sweep is meant to preserve.

We should therefore read the sweep as a compact robustness result, not as a catalogue of completed modules. It asks whether independently trained subjects preserve the same mathematical dependencies when seeds, histories, and local coordinates vary. On that test, the evidence is strong where the world supplies delayed value pressure and the architecture supplies recurrence: state comes to carry information about later consequence, that information affects exposure, and the effect weakens under controls that remove temporal memory or scramble temporal order. The evidence is also meaningful for latent organization, because population trajectories repeatedly become separable by value-relevant structure rather than by arbitrary plotting choices.

But the same sweep also prevents overclaiming. It does not show that every reference role appears with equal strength, and it does not license an identification between learned hidden axes and authored Cave coordinates. Selection-like effects remain partial: cue sensitivity and weighting are present, but an explicit bottleneck-allocation function is not recovered cleanly. The result is therefore a bounded stability claim. Under matched pressure, sufficient capacity tends to produce certain useful trajectory transformations; it does not guarantee the full reference vocabulary. That boundary is part of the result.

14 XIV. Three Levels Of Claim

We distinguish three strengths of claim because the same vocabulary can be supported in importantly different ways. A function may be directly implemented and experimentally recoverable; it may be approached by a compatible architecture when pressure signals are supplied; or it may arise indirectly in a weaker substrate that was never given the role language at all. These are not interchangeable results. They differ in what is installed, what is learned, what is controlled, and what kind of inference is licensed.

The separation matters because it prevents a common overreading. If a reference system exhibits expectation, selection, retention, regulation, and topology, that shows that the vocabulary can be made operational. It does not yet show that those functions emerge elsewhere. If a weakened Cave-compatible system moves toward the reference under pressure, that shows pressure-shaped adaptation within a prepared space. It does not yet show open-ended discovery. If a compact recurrent subject, trained only through delayed exposure consequences, develops hidden-state structure that predicts future outcome and regulates exposure, the claim is stronger but still graded: it concerns functional resemblance under matched pressure, not identity of internal coordinates.

Thus the section below is organized as a ladder. Each rung has its own evidence standard, its own failure modes, and its own permitted conclusion.

Level 1: Reference functions are operational. At the first rung, we are not yet asking whether a foreign substrate discovers the roles on its own. We are asking a narrower and necessary question: can the proposed vocabulary be made into measurable procedures? In the reference Cave setting, the answer is yes. Expectation can be tested through prediction and surprise reduction; selection through preservation of relevant input under bottleneck; value retention through preferential survival of future-relevant state; regulation through changes in later exposure or sampling; and topology through stable regions, transitions, and trajectory organization. Each role can also be weakened or removed, and the corresponding population geometry changes in the expected direction. That is important because the vocabulary is not merely descriptive. It gives us interventions, controls, recovery criteria, and failure signatures.

This first level therefore licenses an operational claim, not an emergence claim. The reference system already contains the relevant role surface, so success here shows that the functions are experimentally tractable: they can be measured, ablated, and recovered. It establishes the measurement language that the later levels will use when the roles are no longer simply given.

Level 2: Pressure shapes compatible architectures. At the second rung, the roles are no longer simply read out from an intact reference system. We weaken or adapt a Cave-compatible architecture and ask whether pressure signals push its behavior toward the reference pattern. The answer is partial and informative. Fixed pressure rules can produce modest movement, often mixed across generated sequences; learned controllers can do better, and ablations reveal which inputs support the gain. But the interpretation remains bounded. The architecture already lives near the Cave update language, so convergence here is best understood as developmental adaptation within a prepared representational space. It shows that pressure can tune an available capacity toward the proposed function. It does not show that an arbitrary substrate invents the role.

Level 3: Weak recurrent subjects can acquire some resembling functions. The third rung removes the authored role surface more strongly. The evolved controller has recurrence and exposure output, but no native expectation variable, no explicit memory trace, no prediction loss, and no named value-retention module. Under delayed value pressure, it nevertheless develops hidden-state structure that carries future-outcome information and regulates later exposure. In the harder dissociation world, this cannot be reduced to frequency counting or a last-seen reflex. Still, the result is graded. Expectation-like value retention, regulation, and latent geometry are strong; selection remains weaker, closer to cue-weight concentration than full attention. Thus the strongest claim is functional resemblance under matched pressure, not uniform emergence of the whole reference vocabulary.

15 XV. Limitations

The strongest reason to state the limitations first is that the paper's argument depends on them. We have not tried to build a large artificial psychology, nor to show that every pressure produces every role in every substrate. The experiments are deliberately small probes. Their purpose is to ask whether a specified pressure, when paired with a specified capacity, can bend population trajectories in the predicted direction and whether that bending disappears under appropriate controls.

This choice gives the results interpretive clarity, but it also narrows their scope. A compact world lets us identify the causal pressure more cleanly; it does not tell us how the same function behaves when many pressures compete. A small recurrent subject makes hidden-state analysis tractable; it does not establish that richer agents would organize themselves in the same way. A controlled dissociation rules out particular cheap strategies; it does not rule out every possible alternative strategy. Likewise, a recovered role

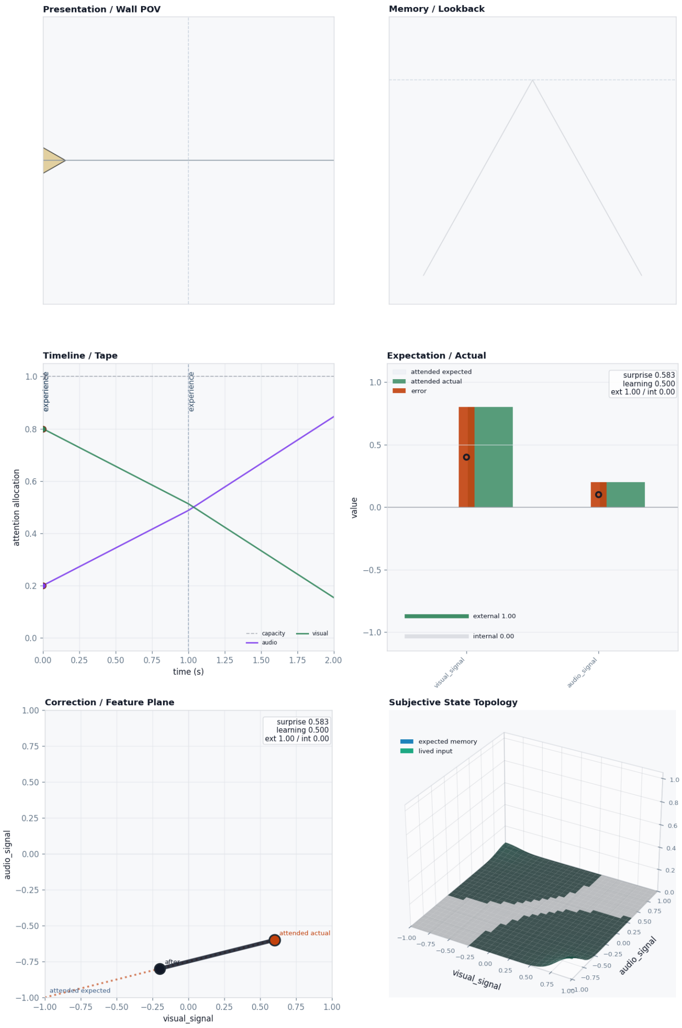

Figure 7: Role-recovery matrix from the result ladder. The ladder separates operational reference roles, pressure-shaped compatible systems, and weaker recurrent subjects that acquire only a subset of resembling functions.

is evidence for a mathematical function under a pressure, not evidence that the system has acquired the full architecture, semantics, or phenomenology associated with that role in the reference vocabulary.

The claims below should therefore be read as bounded claims about pressure-shaped functional resemblance. They are strongest where controls collapse the effect, weaker where the available measurement is only a proxy, and silent where the experiment has not yet supplied the relevant pressure, capacity, or comparison class.

Several limitations follow directly from the experimental design. First, the evolved worlds remain compact. Even in the harder dissociation setting, future outcome and future value are not fully disentangled across the space of possible environments; the experiment separates enough to defeat the targeted cheap strategies, not enough to exhaust the problem. Second, our evidence for selection is deliberately weaker than our evidence for value retention and regulation. In the recurrent subjects, selection is currently measured through input-weight concentration and sensitivity patterns. That is evidence for cue weighting, but it is not yet evidence for a full dynamic attention mechanism that allocates access over time under changing task demands.

Third, cross-architecture comparisons must remain functional comparisons. When a learned hidden state supports future-outcome decoding or exposure regulation, it resembles the reference role by what it does, not because any hidden coordinate is identical to a Cave feature axis. Fourth, the sweeps are compact. They test robustness over a local range of architectures, seeds, and generated worlds; they do not establish a universal scaling law. Finally, these systems are not biological organisms. The results concern minimal recurrent controllers under specified pressures, and should not be read as claims about neural implementation, development, affect, or lived meaning.

Most importantly, none of these results establishes subjective experience, consciousness, or phenomenal feeling. The term "subjective" in this paper is used in the restricted sense introduced in Paper 1: a trajectory as it is conditioned by a system's own retained state, access limits, and update rules. It is not a claim that the system has a point of view in the ordinary psychological or phenomenological sense.

The strongest claim we can honestly draw is narrower and, we think, more useful. In compact delayed value-control environments, weak recurrent subjects can be selected to carry future-relevant information in latent state, to retain that information in a value-shaped way, to use it to regulate later exposure, and to form a recoverable geometry over their hidden trajectories. These properties are not read from authored Cave variables and are not supplied as named modules. They appear as measurable functions of the learned dynamics, and they weaken or collapse under the controls that remove recurrence, disrupt temporal structure, or break the relevant pressure-capacity pairing.

Thus the paper supports a bounded emergence claim: pressure can shape capable minimal systems toward recognizable mathematical roles. It does not license a phenomenological claim, a biological claim, or a claim of complete role recovery.

16 XVI. Remaining Work

The remaining work is to turn the population-trajectory argument into a single cross-substrate design. In the present evidence, several pieces are already separated: same-pressure convergence, matched-start treatment separation, control collapse, and evolved-subject regulation. The next step is to place them in one factorial frame,

$$\text{treatments} \times \text{starts or seeds} \times \text{substrate conditions}.$$

This design would let us ask a sharper question: not merely whether one system exhibits a role-like behavior, but whether matched pressures deform populations of trajectories in comparable ways across different implementational capacities.

In such a design, each treatment would define a pressure protocol; each start or seed would define an initial subjective or latent condition; and each substrate condition would define the available capacity. The primary object would be the aligned family of trajectories, not an isolated performance score. We would therefore measure whether shared treatment produces convergence across starts, whether matched starts separate by treatment, whether negative controls collapse the relevant geometry, and whether interventions remove only the structure that the proposed role requires.

This would also make the comparison between reference systems and learned systems more disciplined. We would not ask whether their coordinates match. We would ask whether their trajectory transformations, failures, and recoveries occupy the same functional position under matched pressure.

The condition axis should therefore be deliberately broad. It should include the authored Cave baseline, weakened Cave variants, adaptive Cave-compatible systems, compact evolved recurrent subjects, and the relevant negative controls: hidden-state reset, non-recurrent controls, memory removal, attention removal, prediction removal, and temporal shuffling. Each condition should receive matched treatment protocols and matched starts wherever the substrate permits them. The point is not to force identical coordinates across systems, but to ask which capacities are sufficient for the same pressure to produce the same kind of trajectory deformation, and which removals destroy it.

The world axis should also become more adversarial to cheap substitutes. Cue reliability should vary, delays should be stochastic, distractors should appear during the delay interval, and cue meanings should sometimes reverse. Relevance should depend on context rather than on a fixed cue identity, and the subject should sometimes choose among several exposure or action options rather than a single scalar approach–avoid output. These manipulations separate recurrence from mere persistence, value retention from frequency counting, selection from raw salience, and regulation from a fixed motor bias. In that setting, a role-like result would have to survive not only ordinary variation, but targeted alternatives designed to imitate it.

The frequency–value dissociation therefore changes status. It is no longer merely an item on the wish list, but a calibration result that should be generalized. The present harder world shows that an evolved recurrent subject can retain outcome-relevant sign when frequency and value are pulled apart. The remaining question is how stable that result is when the world family becomes richer, less regular, and more hostile to simple policies.

The next design should treat dissociation as a generative principle rather than as a single benchmark. Base rate, reward sign, reward magnitude, delay length, distractor density, contextual cue meaning, and action opportunity should vary independently across generated worlds. Some worlds should reward rare cues, some should punish common cues, some should reverse meanings by context, and some should make the relevant feature depend on a preceding latent state. We can then ask whether the same pressure continues to produce future-outcome decodability, sign- or magnitude-sensitive retention, regulated exposure, and stable latent geometry.

This would also sharpen the negative result space. If a subject succeeds only when frequency, recency, and value remain partially aligned, then the claimed function is narrow. If it succeeds across dissociated world families and fails under the predicted capacity removals, then the resemblance claim becomes correspondingly stronger.

17 XVII. Conclusion

We can now state the central conclusion of the pressure program. Across the result ladder, Cave's Episode contract gives us a common measurement surface for asking when a trajectory transform is present, weakened, removed, or recovered. At the first level, the reference roles are not merely labels: expectation, selection, value retention, regulation, and topology can be operationalized through reports, controls, and ablations. At the second level, Cave-compatible systems move toward those roles when pressure signals make the corresponding transformations useful. At the third level, compact recurrent subjects that were not given the Cave roles nevertheless acquire some analogous functions under delayed value and exposure-control pressure.

The common lesson is the pressure/capacity/function thesis. A pressure explains why a transformation would be useful; a capacity explains how any substrate could implement it. When either side is missing, the function does not appear in the same way. Delayed consequence does not produce memory in a system that cannot carry state. Bottleneck pressure does not produce selection in a system that cannot update selectively. Conversely, capacity without pressure gives us possible computation, but not a reason for that computation to stabilize. The evidence therefore supports a graded computational account: recurring pressures can make certain mathematical roles worth learning, while substrate capacity determines which roles can actually be learned.

For that reason, the conclusion should be read with deliberate restraint. We have not shown that any artificial system is conscious, nor that the presence of these roles is sufficient for consciousness. The measurements in this paper do not settle phenomenal status, intrinsic perspective, or moral standing. They show something narrower and, we think, more tractable: functions that are often invoked in descriptions of subject-like experience can be specified as trajectory transformations, placed under pressure, removed by controls, and compared across substrates without assuming a shared internal coordinate system.

This narrowing is important. If expectation-like state, value-shaped retention, regulation, and latent organization recur under matched pressures, the recurrence does not license a metaphysical conclusion by itself. It licenses a computational one. These functions look less like arbitrary names assigned after the fact and more like useful mathematical responses to recurrent structural problems: delay, bottleneck, future consequence, controlled exposure, and repeated geometry. The stronger the capacity of the substrate to retain, update, couple, and act on state, the more available these responses become; the weaker the capacity, the more the same pressure produces collapse or absence rather than role formation.

Thus the final thesis is conditional rather than magical. Pressure supplies a gradient of usefulness, not a mechanism of creation. It can make memory, selection, or regulation worth having; it cannot conjure recurrence, writable state, selective update, or action coupling where the substrate lacks them. The negative controls are therefore not embarrassments at the edge of the theory. They are part of the theory's content: when the relevant capacity is removed, the pressure remains, but the role collapses or changes form.

The positive result is the converse. Where a substrate can retain state, update it selectively, couple it to value, and let it influence future sampling, the same families of trajectory transformations become stable solutions to the same families of problems. We should not expect coordinate identity across systems, and we should not expect all roles to emerge with equal strength.

But we should expect intelligible functional resemblance: preserved future-relevant state, reduced surprise, regulated exposure, separable population geometry, and predictable failures under matched controls. That is the level at which the Cave vocabulary earns its use here. It is not a catalogue of installed modules. It is a map of mathematical roles that pressure can favor when capacity makes them implementable.